

NURSES CALLING SYSTEM

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<http://20.163.9.187:3000/> - FRONTEND URL

<http://20.163.9.187:5001/> - BACKEND URL

EMAIL : admin@carecall.com

PASSWORD: [Admin@123](#) (super_admin)

Project Overview:

The **Nurses Calling System** is a communication system used in hospitals that allows patients to call nurses quickly when they need assistance. The system improves patient care by providing an efficient way for patients to request help without leaving their bed.

The system usually consists of **call buttons placed near patient beds**, a **central monitoring unit**, and **notification systems** such as lights, buzzers, or display panels.

Objective:

- To provide **quick communication** between patients and nurses.
- To improve **patient safety and response time**.
- To reduce **manual monitoring workload** for nurses.
- To provide **real-time alerts** when patients require help.

Scope of the Project:

The Nurses Calling System can be used in:

- Hospitals
- Clinics
- Elderly care homes
- Rehabilitation centers
- ICU wards

The system can be implemented using **embedded systems, IoT, or software-based monitoring systems.**

System Overview:

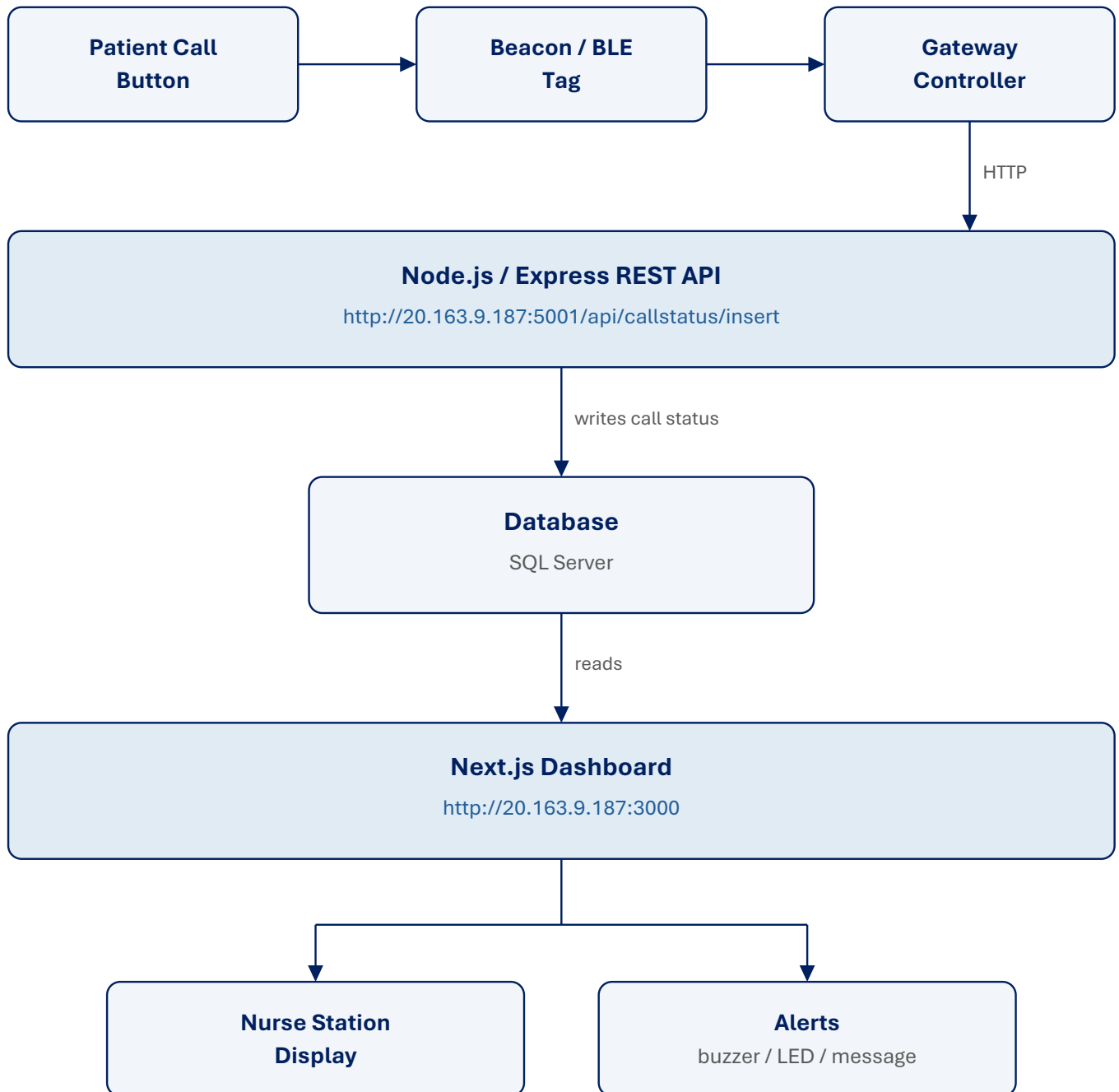
The system works when a patient presses a **call button**. The signal is sent to a **central controller or server**, which then alerts the **nursing station** through:

- Visual indicators
- Alarm sounds
- Display panels
- Mobile notifications

System Architecture:

1. Components of the system:
2. Patient Call Unit
3. Central Controller
4. Nurse Station Display
5. Alert System
6. Database (optional for logging)

System Architecture:



Tables: call status, rooms, users. Status values: 1 = CALL, 0 = MUTE, 2 = RESET

Patient Button → Microcontroller → Server → Nurse Station Display



Database

Software Requirements:

Software used for development:

- Backend: Node.js
- Database: SQL Server
- Frontend: React.js

Functional Modules:

1. Patient Module:

- Patient presses call button
- Signal sent to system
- Request is logged

2. Nurse Station Module:

- Displays patient room number
- Shows active calls
- Allows nurse to acknowledge the request

3. Alert System:

- Sound alarm
- LED indicator
- Display message

4. Admin Module

- Manage users and rooms
- Monitor call history
- Generate reports

System Workflow:

- Patient presses the **call button**
- Signal is sent to the **controller**
- System identifies **patient room**
- Notification is sent to **nurse station**
- Nurse acknowledges the call
- System updates the **call status**

Advantages:

- Faster response time
- Improved patient care
- Reduced nurse workload
- Real-time monitoring
- Emergency alerts

DETAILS:

OID= ORG001 (organization id)

NURSES CALLING string:

http://20.163.9.187:5001/api/callstatus/insert
orgId=00003&hid=2408202601&beacon&r03=3

beacon string:

orgId=00003&hid=2408202601&beacon&

Send data --> Service --> Database --> UI

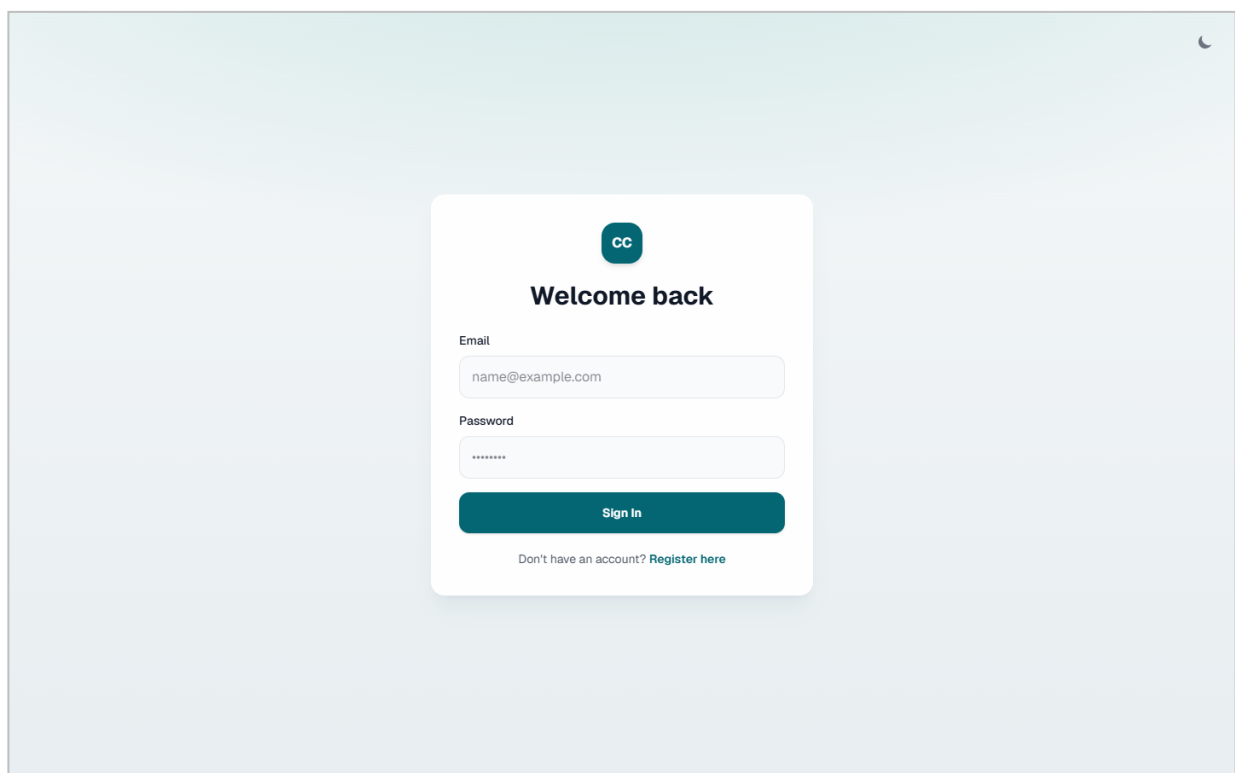
PARAMETER DETAILS:

dnum	ROOM NO.
floor	FLOOR NO.
status	STATUS => 1-CALL, 0-MUTE, 2-RESET

Application Screenshots:

1. Login Page

Users sign in with their registered email and password.



2. Dashboard (User 1 - AKG Hospital)

Active calls, call statistics, recent activity and quick actions.

CC

Care Call

Dashboard

Reports

Settings

Logout

Active Calls (0 active)

Organisation: AKG HOSPITAL (00002)

No active calls

Total Calls

0

Pending Calls

0

Resolved Today

18

Response Time

0.7 min

Recent Activity

View all

Call from 462 Floor 1
57 minutes ago

Normal Resolved
Reset: 22-09-2026 02:40:44 PM
22-09-2026 02:39:52 PM

Call from 467 Floor 1
78 minutes ago

Normal Resolved
Reset: 22-09-2026 02:18:59 PM
22-09-2026 02:18:37 PM

Call from 467 Floor 1
109 minutes ago

Normal Resolved
Reset: 22-09-2026 01:48:29 PM
22-09-2026 01:47:44 PM

Quick Actions

View Reports

Settings

3. Dashboard (User 2 - St. Martin's Hospital)

Floor/HID selector with call activity for the selected devices.

CC

Care Call

Dashboard

Reports

Settings

Logout

SELECT FLOOR/HID

All devices

Top — All devices (0 active)

Organisation: ST. MARTIN'S HOSPITAL, AMBALAVAYAL (00003)

No active calls for All devices

Total Calls

0

Pending Calls

0

Resolved Today

0

Response Time

-

Recent Activity

View all

Call from 703 Floor 7
25845 minutes ago

Normal Beacon
Reset: 10-09-2026 11:57:24 AM
04-09-2026 04:52:21 PM

Call from 704 Floor 7
25845 minutes ago

Emergency Beacon
Reset: 10-09-2026 11:57:24 AM
04-09-2026 04:52:16 PM

Call from 702 Floor 7
25845 minutes ago

Emergency Beacon
Reset: 10-09-2026 11:57:24 AM

Quick Actions

View Reports

Settings

4. Reports

Call history with filters, summary reports and Excel/PDF export.

CC

Care Call

Dashboard

Reports

Settings

Logout

Search by Room or Call ID

All Statuses

All

All Rooms

Start Date

dd-mm-yyyy

End Date

dd-mm-yyyy

Export Excel

Export PDF

Summary Reports

Calls per Day

Calls per Room

Attending Lag

Beacon Data

Call History

Showing 1-10 of 2357

Rows per page: 10

#	ROOM	DEPARTMENT	HID	FLOOR	CALL TYPE	STATUS	MUTED	CREATED	MUTED AT	RESET AT
1	462	Rooms	1234567891	1	Normal	Resolved	Unmuted	22-09-2026 02:39:52 PM	22-09-2026 02:39:52 PM	22-09-2026 02:39:52 PM
2	467	Rooms	1234567891	1	Normal	Resolved	Unmuted	22-09-2026 02:18:37 PM	22-09-2026 02:18:37 PM	22-09-2026 02:18:37 PM
3	467	Rooms	1234567891	1	Normal	Resolved	Unmuted	22-09-2026 01:47:44 PM	22-09-2026 01:47:44 PM	22-09-2026 01:47:44 PM
4	468	Rooms	1234567891	1	Emergency	Resolved	Unmuted	22-09-2026 12:46:27 PM	22-09-2026 12:46:27 PM	22-09-2026 12:46:27 PM
5	461	Rooms	1234567891	1	Normal	Resolved	Unmuted	22-09-2026 11:42:37 AM	22-09-2026 11:42:37 AM	22-09-2026 11:42:37 AM

5. Calls per Day Report

Daily call counts with filters and PDF download.

CC

Care Call

Dashboard

Reports

Settings

Logout

Back to Reports

Search by Room or Call ID

All Statuses

All

All Rooms

Start Date

dd-mm-yyyy

End Date

dd-mm-yyyy

Daily call counts

2358 total calls

Download PDF

DATE	COUNT
10-07-2026	39
11-07-2026	4
13-07-2026	28
14-07-2026	115
15-07-2026	64
16-07-2026	8
17-07-2026	4
18-07-2026	48

6. Calls per Room Report

Number of calls raised from each room.

CC

Care Call

Dashboard

Reports

Settings

Logout

Calls per Room

Back to Reports

Search by Room or Call ID

All Statuses

All

All Rooms

Start Date

dd-mm-yyyy

End Date

dd-mm-yyyy

Calls by room

2358 total calls

Download PDF

ROOM	COUNT
462	273
468	273
463	212
461	193
459	150
451	142
464	132
457	130

7. Attending Lag Report

Calls whose response time exceeded the configured lag (minutes).

CC

Care Call

Dashboard

Reports

Settings

Logout

Attending Lag

Back to Reports

Search by Room or Call ID

All Statuses

All

All Rooms

Start Date

dd-mm-yyyy

End Date

dd-mm-yyyy

Attending Lag (min)

15

Attending Lag > 15 minutes

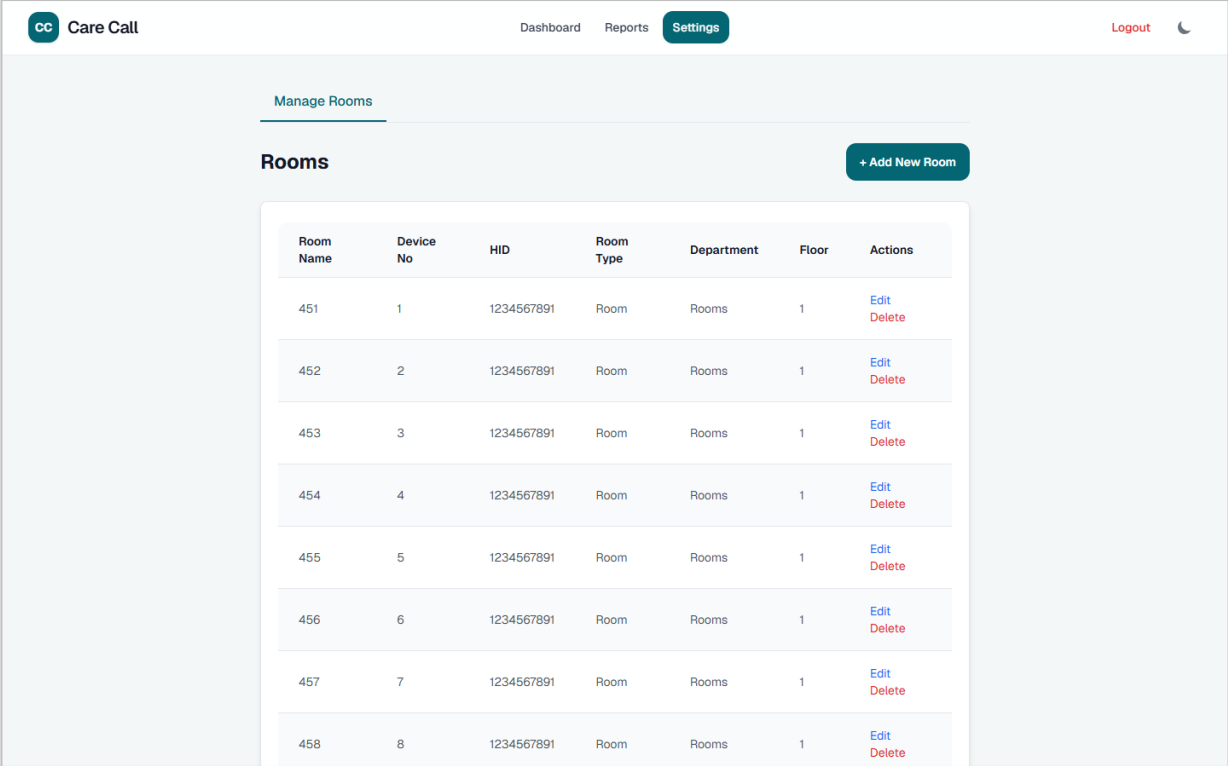
101 calls

Download PDF

CALL ID	ROOM	CREATED	RESET	LAG (MIN)
CALL_1784811699035_rpbuh7	459	23-07-2026 06:31:39 PM	30-07-2026 05:29:42 PM	10018
CALL_1784023810851_yayzm5	461	14-07-2026 03:40:10 PM	20-07-2026 02:57:26 PM	8597
CALL_1784021940952_wtg7j9	459	14-07-2026 03:09:00 PM	20-07-2026 12:14:24 PM	8465
CALL_1787648918797_1lwOxm	459	25-08-2026 02:38:38 PM	29-08-2026 09:52:34 PM	6193
CALL_1785412825591_hzfxao	452	30-07-2026 05:30:25 PM	03-08-2026 12:01:59 PM	5431
CALL_1789748751522_jkk4z5	458	18-09-2026 09:55:51 PM	21-09-2026 09:23:20 PM	4287
CALL_1785484042940_aqrl4m	453	31-07-2026 01:17:22 PM	03-08-2026 12:02:00 PM	4244

8. Settings - Manage Rooms

Add, edit and delete rooms with device number, HID, type and floor.



The screenshot displays the 'Manage Rooms' section of the 'Care Call' application. The top navigation bar includes 'CC Care Call', 'Dashboard', 'Reports', and 'Settings' (which is active). A 'Logout' link and a moon icon are on the right. Below the navigation bar, the 'Manage Rooms' section is titled. A '+ Add New Room' button is located in the top right corner of the room management area. The main content is a table with the following data:

Room Name	Device No	HID	Room Type	Department	Floor	Actions
451	1	1234567891	Room	Rooms	1	Edit Delete
452	2	1234567891	Room	Rooms	1	Edit Delete
453	3	1234567891	Room	Rooms	1	Edit Delete
454	4	1234567891	Room	Rooms	1	Edit Delete
455	5	1234567891	Room	Rooms	1	Edit Delete
456	6	1234567891	Room	Rooms	1	Edit Delete
457	7	1234567891	Room	Rooms	1	Edit Delete
458	8	1234567891	Room	Rooms	1	Edit Delete

Future Enhancements:

- Mobile app notifications for nurses
- Integration with hospital management system
- Voice communication
- AI-based emergency detection
- Wireless IoT sensors

Conclusion:

The **Nurses Calling System** helps improve communication between patients and healthcare staff. It enhances hospital efficiency and ensures timely medical assistance. Implementing such systems leads to better patient satisfaction and improved healthcare service.